

RL Portfolio Optimization v2

LSTM Backbone + Bug Fixes

Assets: AAPL | MSFT | GOOGL | AMZN | TSLA

Train: 2018-2022 | Val: 2023 | Test: 2024

Test Results (2024)

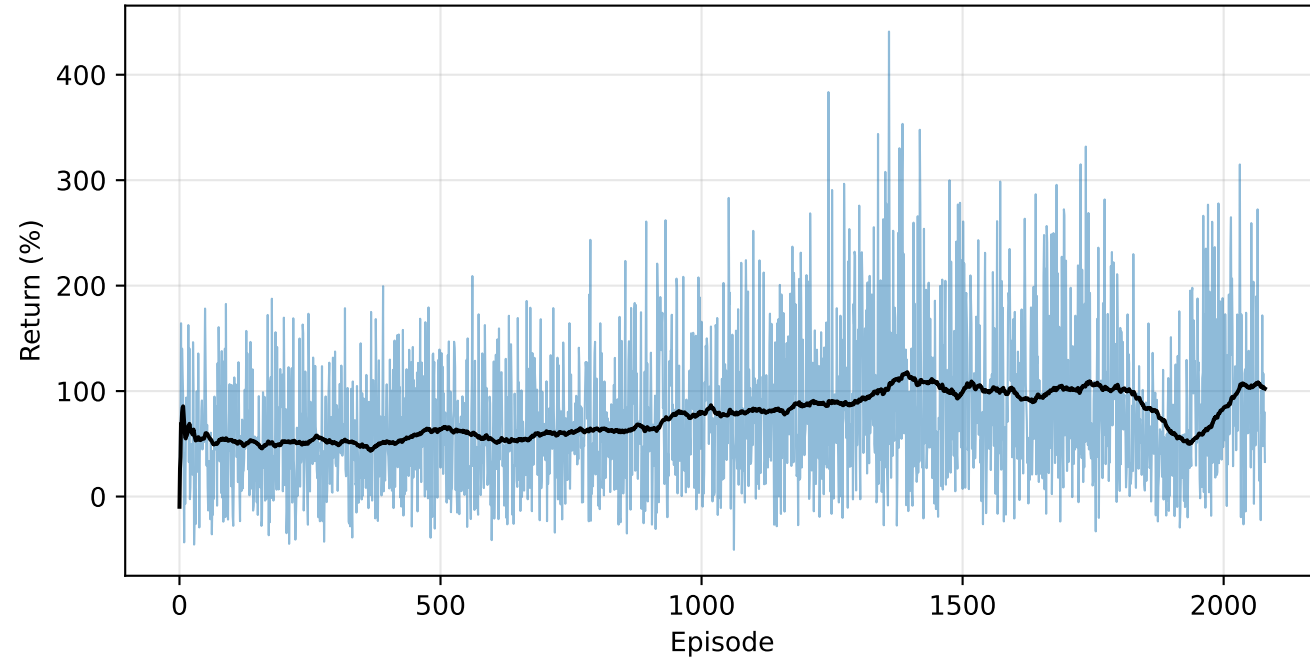
Experiment	Cum.Return	Sharpe	MaxDrawdown	CVaR_0.95	Turnover
PPO_CVaR	+35.15%	1.421	-14.21%	-0.0282	0.0161
PPO_Var	+37.27%	1.479	-14.71%	-0.0287	0.0161
TD3_CVaR	+19.82%	0.639	-15.58%	-0.0405	0.3063
TD3_Var	+33.58%	1.373	-14.37%	-0.0283	0.0969
EqualWeight	+45.44%	1.514	-17.32%	-0.0344	0.0000

Cumulative Return — Test Set (2024)

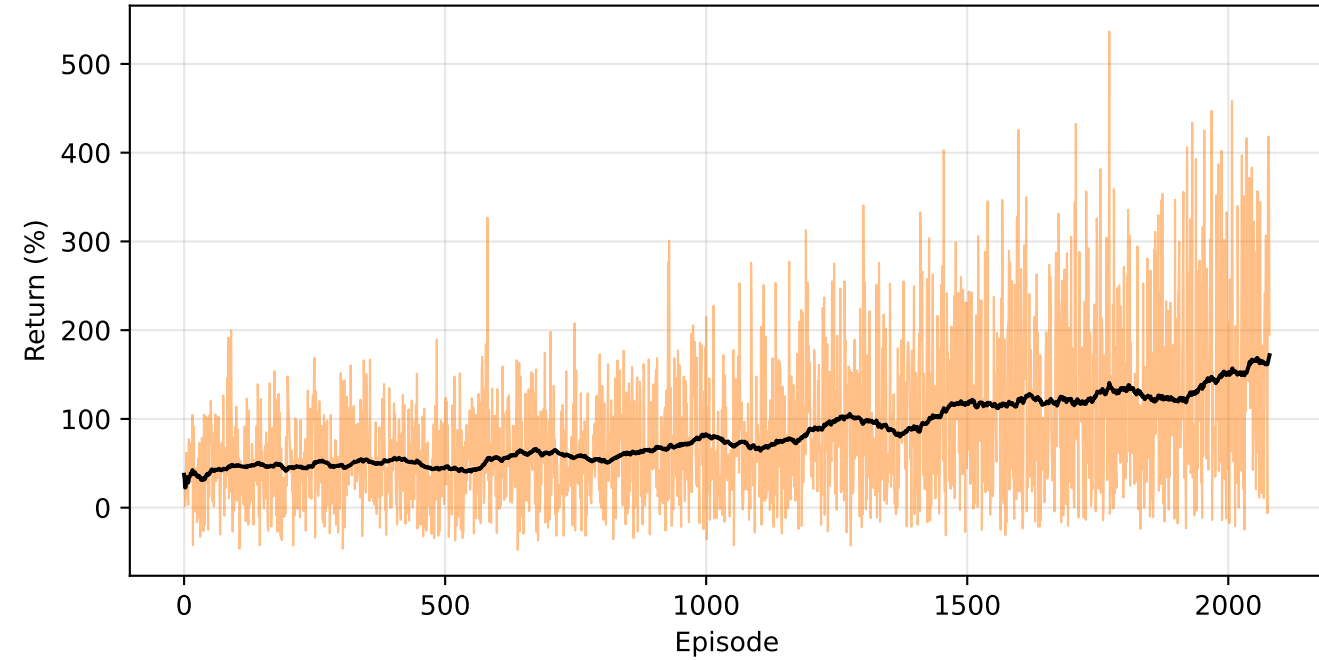


Training Curves

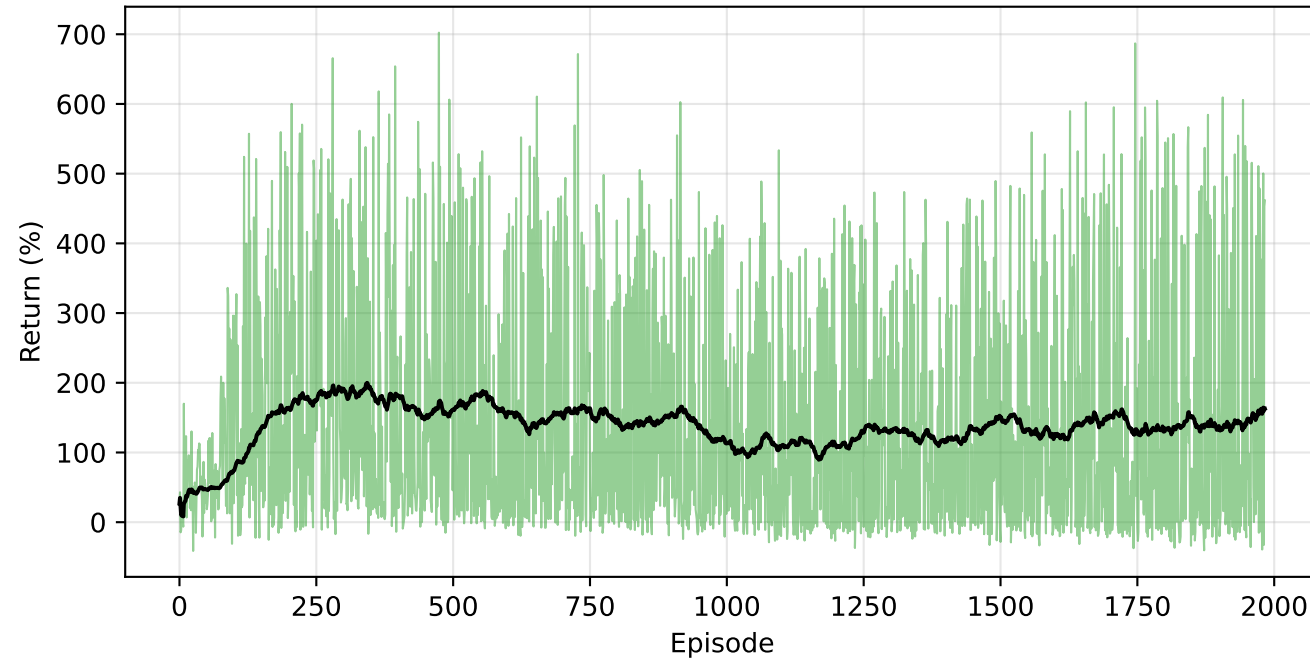
PPO_CVaR Training



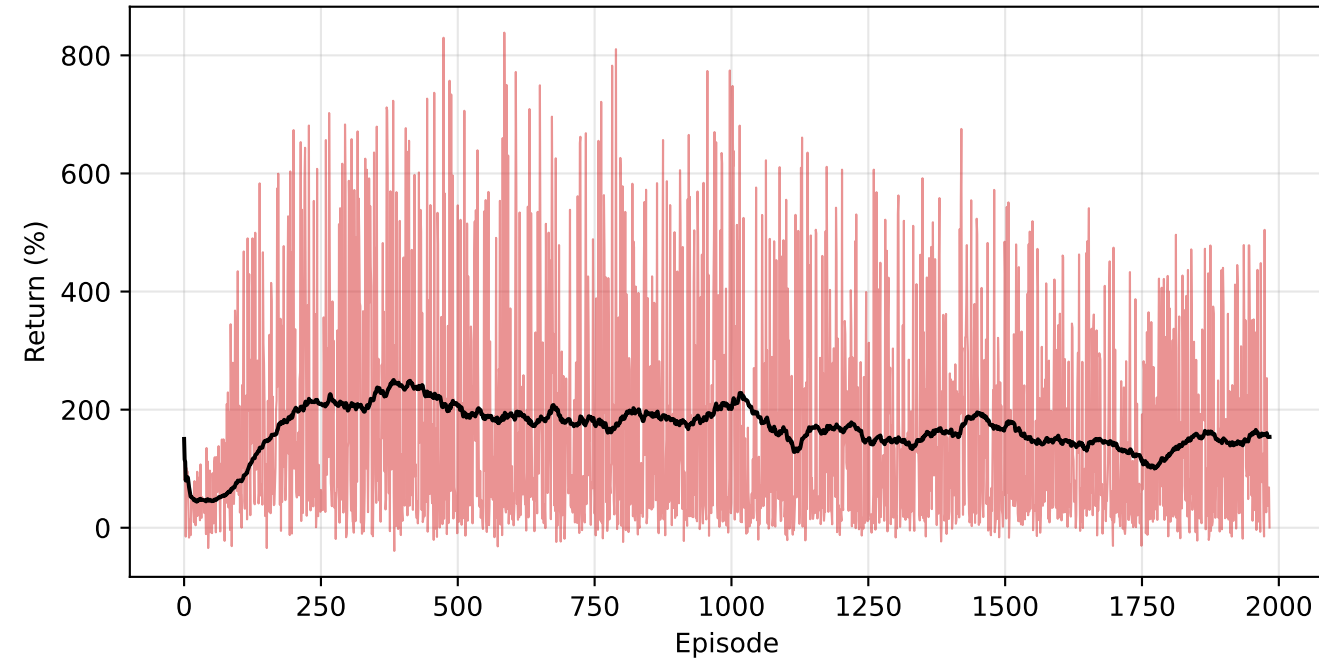
PPO_Var Training



TD3_CVaR Training

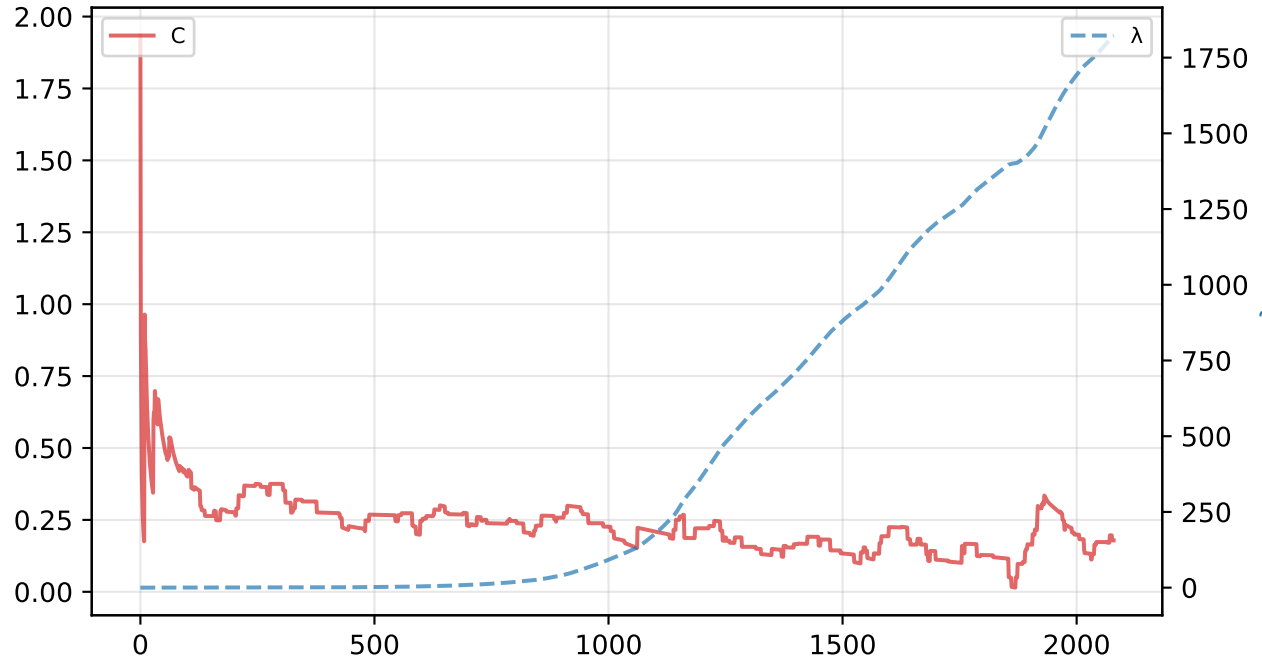


TD3_Var Training

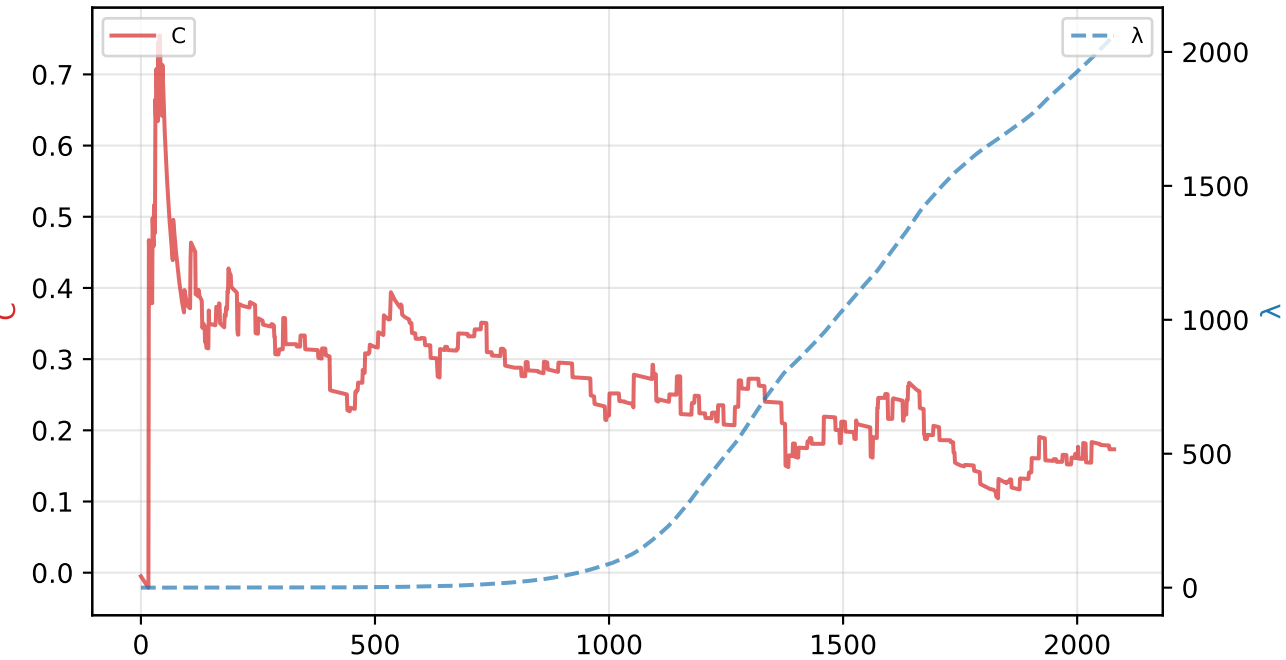


CVaR Violation and λ

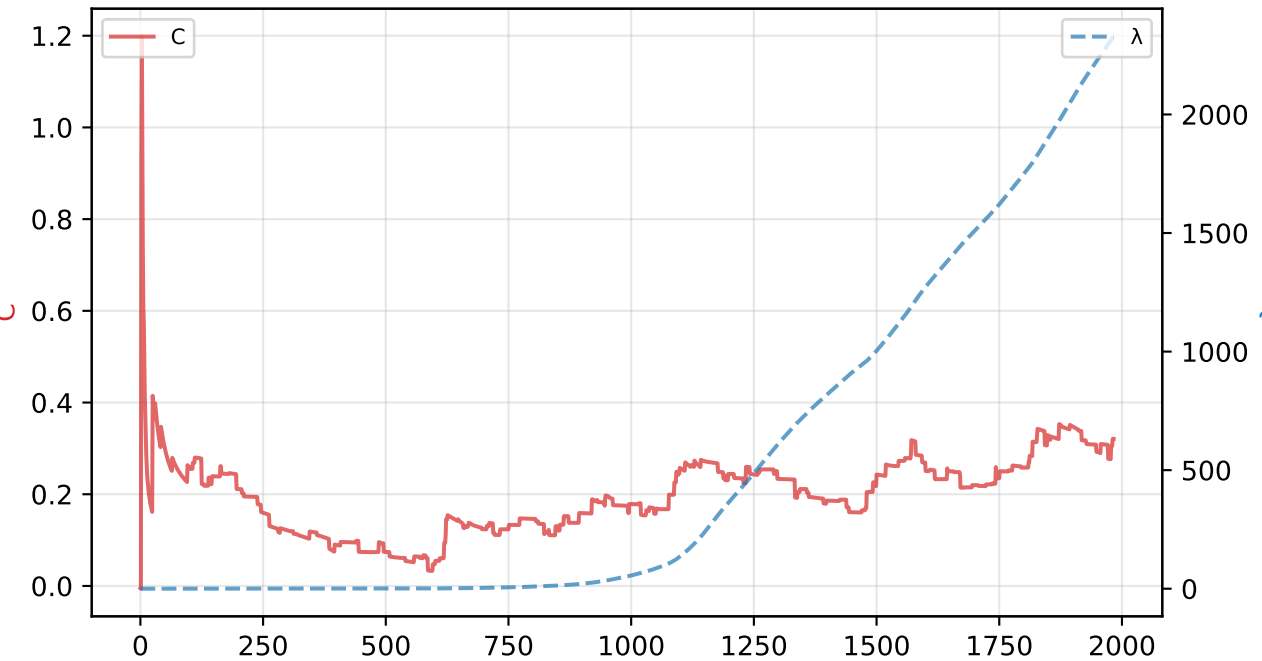
PPO_CVaR Constraint



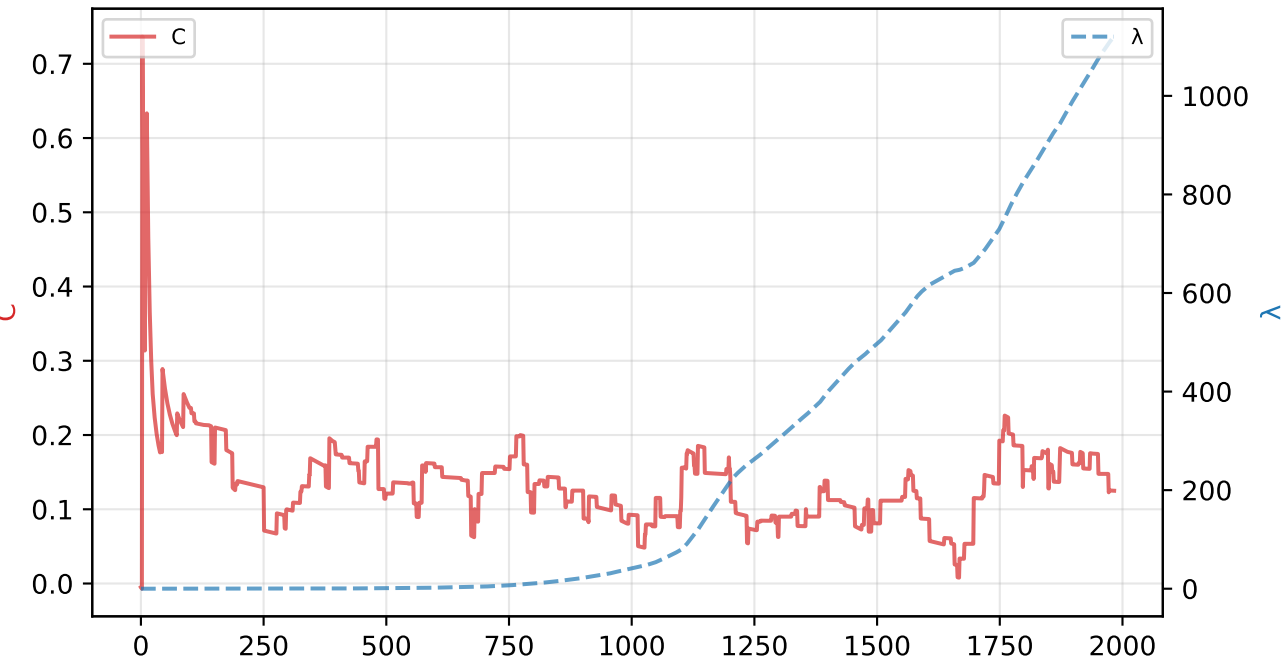
PPO_Var Constraint



TD3_CVaR Constraint

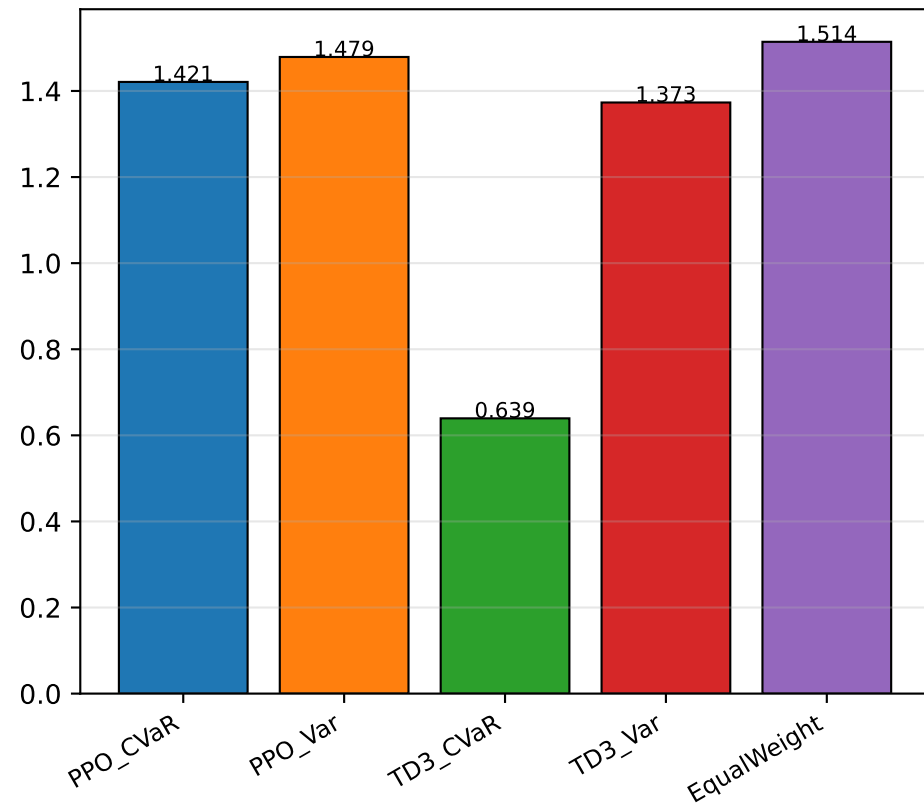


TD3_Var Constraint

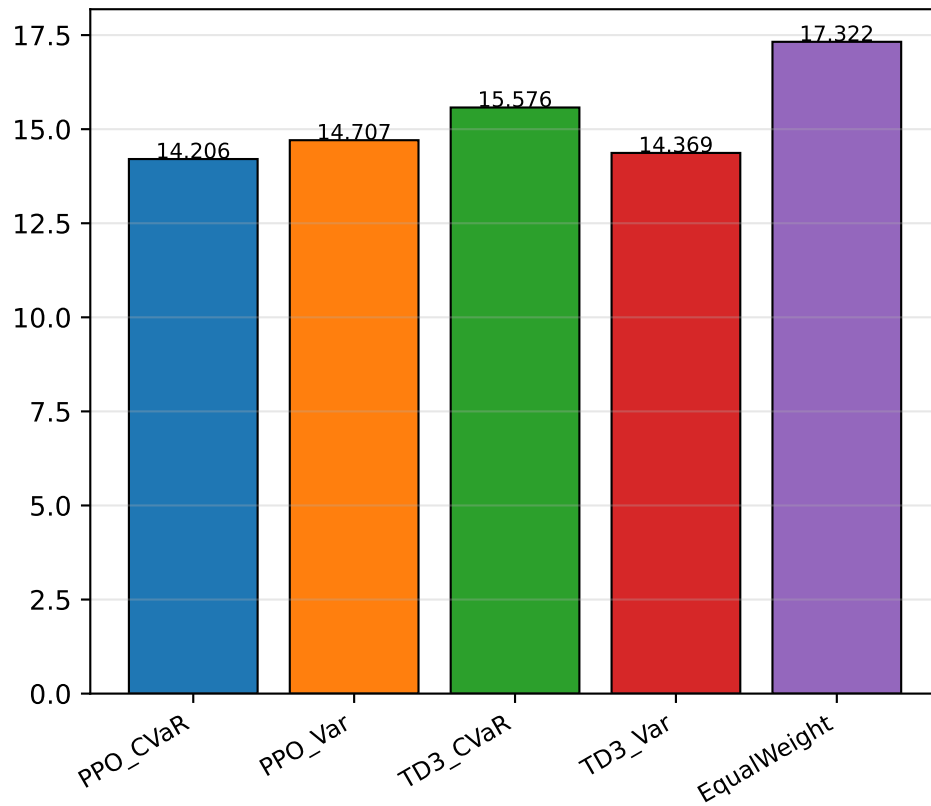


Test Set Metrics

Sharpe Ratio



Max Drawdown (%)



Avg Turnover

